COMPOSITION ISOMORPHISM FOR THE UPPER SHRIEK OF LOCALLY QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Dual composition isomorphism for the lower shriek of separated locally quasi-finite morphisms: a canonical isomorphism f_! composed with g_! related to (g composed with f)_!, with the corresponding commutative diagram for three morphisms (more-etale, Lemma lemma-lqf-shriek-composition, canonical tag 0F6T). cited at proof line 2490. The single named input: the dual lower-shriek statement into which the claim translates by uniqueness of adjoints.

Source reference. Original lemma 0F6T: sources/stacks-chow/tags.txt, source lines 18983–18983. Cited at proof line 2490: the corresponding dual statement for the f_! functors from which the claim follows.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to Z$ be separated and locally quasi-finite morphisms. There is a canonical isomorphism $(g \circ f)^! \to f^! \circ g^!$. Given a third locally quasi-finite morphism $h : Z \to T$ the diagram*

$$\begin{array}{ccc} (h \circ g \circ f)^! & \longrightarrow & f^! \circ (h \circ g)^! \\ \downarrow & & \downarrow \\ (g \circ f)^! \circ h^! & \longrightarrow & f^! \circ g^! \circ h^! \end{array}$$

commutes.

Proof. By uniqueness of adjoint functors, this immediately translates into the corresponding (dual) statement for the functors $f_!$. See Lemma 0F6T. □

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